From Controversy to Consensus: Modeling Community-Sensitive Common-Ground Management in German Discourse Markers

Considering the sentences in (**??**):

(1) Professor Weber kündigt (sofern ich weiß / wie du ja weißt / wie wir wissen / ja / bekanntlich) nächste Woche, deshalb sollten wir eine Abschiedfeier planen.
Professor Weber resigns as.far.as I know / as you PRT know / as we know / PRT / well.known next week therefore should we a farewell.party plan
'Professor Weber is resigning next week (as far as I know / as you know / as we know / as you know / as is well known); therefore we should plan a farewell party.'

The expressions in parentheses form a scale with respect to the degree to which the antecedent proposition p is presented as shared and settled in the relevant epistemic community, ranging from weak epistemic limitation (*sofern ich weiß*) to strong public accessibility (*bekanntlich*). Intuitively, as listeners, we can sense systematic differences concerning both the antecedent p and the consequent q. First, there is a gradient hierarchy of sharedness of p depending on the marker used, ranging from a situation in which only a few people know the news to one in which almost everyone in the relevant community is assumed to know it. The markers thus differ in how strongly they present p as established within a shared epistemic space. Second, this hierarchy is context-dependent in two ways. On the one hand, properties of the proposition itself matter: if Professor Weber is a widely known public figure, p is more likely to count as broadly accessible information. On the other hand, the relevant epistemic community must be determined: the denotation of *wir* and the assumed overlap between speaker and addressee crucially shape the felicity of stronger markers. If the speaker has reason to assume that the addressee shares the same epistemic background—e.g., because they are colleagues—he may use a marker high on the scale even if Professor Weber is not widely known outside that community. Sharedness is therefore evaluated relative to a contextually determined epistemic group. Third, there is a gradient hierarchy of speaker commitment with respect to q. The speaker may genuinely aim to convince others to plan the farewell party. The more strongly p is presented as shared and uncontroversial, the more forcefully it can function as a premise for q. In this sense, marker choice reflects not only assumptions about common ground, but also the speaker's own belief strength and argumentative goal.

This study focuses on a class of German discourse markers — *ja*, *wie wir wissen* ('as we know'), and *bekanntlich* ('as is well known') — which we group under the label *consensus markers*. Previous accounts have largely concentrated on the particle *ja*. It is commonly assumed not to contribute to the truth-conditional content of p (Gutzmann, 2015; Grosz, 2021), but to operate at the discourse level by presenting p as not up for dispute. Accordingly, *ja* has been described as marking p as KNOWN (Thurmair, 1989), as indicating (speaker-assumed) common knowledge (Karagjosova, 2004), or as conveying that p is uncontroversial (Jacobs, 1991; Zimmermann, 2012; Waltereit, 2001). While these formulations converge on the intuition that p is treated as settled, they are typically categorical rather than gradient and do not provide a basis for quantitative predictions. At the same time, it has been observed that *ja* can be used as common ground management tool to strategically support broader communicative goals (Repp, 2013), the so-called "manipulative use" (Döring, 2016). By presenting p as uncontroversial, the speaker may facilitate discourse continuation and promote stability in the evolving discourse structure. Recent experimental work

(Seemann, 2025) further suggests that the use of *ja* is closely tied to causal reasoning contexts, where p functions as a premise for subsequent argumentative moves.

Building on these insights, we argue that two aspects have not yet been sufficiently integrated into a unified formal account. First, notions such as sharedness and controversy are better understood as gradient rather than hard cut-off properties. A proposition may be more or less widely shared within a community, and more or less controversial, depending on background assumptions and world knowledge. Second, the use of consensus markers crucially involves theory-of-mind reasoning: speakers must estimate the epistemic states of their interlocutors, the distribution of beliefs within a relevant community, and their own degree of commitment to an argumentative goal as in our running example (**??**) (e.g. convincing others to accept q on the basis of p). In this study, we propose a formal framework within probabilistic pragmatics (Goodman and Frank, 2016; Degen, 2023; Franke and Jäger, 2016) that models consensus markers as encoding thresholded constraints over a utility function. This function integrates graded common-ground alignment, community-relative perceived controversy, and goal-directed speaker commitment, thereby yielding explicit and testable predictions about marker choice across contexts.

Comprehension model. On the comprehension side, the listener observes an utterance u containing a consensus marker applied to a proposition p, followed by a proposed action q. The listener performs a joint inference about (i) the speaker's persuasive goal strength g in advancing q, and (ii) the perceived controversy pc of p, and, as a downstream consequence of this inference, about whether to adopt Adopt(q).

We treat perceived controversy as community-relative and gradient, and decompose it into two components: $pc_{\text{prop}} \in [0, 1]$ (propositional controversy; e.g. implausibility relative to world knowledge) and $pc_{\text{prag}} \in [0, 1]$ (pragmatic controversy; e.g. expected disagreement / lack of common-ground alignment in the relevant epistemic community). Overall perceived controversy is computed by an interaction function

$$pc \;=\; PC(pc_{\text{prop}}, pc_{\text{prag}}) \;=\; pc_{\text{prop}} \cdot pc_{\text{prag}}, \tag{1}$$

so that high propositional controversy can be attenuated by low pragmatic controversy (and vice versa), reflecting the idea that controversy is assessed relative to an epistemic community rather than absolute plausibility alone.

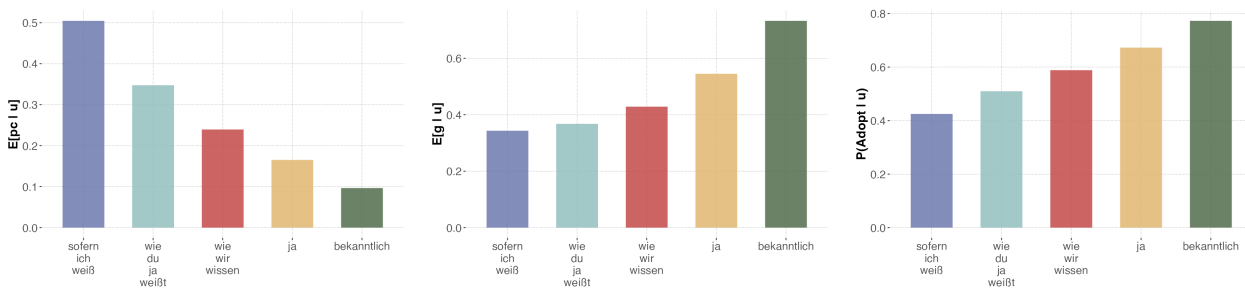
Consensus markers lexicalize thresholds over a utility function:

$$U(p; pc, g) = -w_{pc} \cdot pc + w_g \cdot g - w_{int} \cdot (pc \cdot g), \tag{2}$$

where $w_{pc}, w_g, w_{int} > 0$. The interaction term captures the idea that persuasive effort is increasingly costly when perceived controversy is high.

Each marker u_i is associated with a lower threshold θ_i and an upper threshold θ_{i+1} (the lower threshold of the next stronger marker), with $\theta_{\text{last+1}} = +\infty$ for the strongest marker. A marker is felicitous when the utility falls within its licensed interval $[\theta_i, \theta_{i+1})$, formalised as a soft interval likelihood:

$$P(u_i \mid pc, g) \;\propto\; \sigma\big(\lambda\,(U(p; pc, g) - \theta_i)\big) \;\cdot\; \sigma\big(-\lambda\,(U(p; pc, g) - \theta_{i+1})\big), \tag{3}$$



(a) Posterior expected perceived controversy $\mathbb{E}[pc \mid u]$ per marker.

(b) Posterior expected persuasive goal strength $\mathbb{E}[g \mid u]$ per marker.

(c) Predicted action adoption probability $P(\text{Adopt}(q) \mid u)$ per marker.

Figure 1

where $\sigma(x) = (1 + e^{-x})^{-1}$ is the logistic function and $\lambda > 0$ controls the sharpness of the interval boundaries. The second factor is omitted (set to 1) for the strongest marker. This formulation ensures that each (pc, g) point preferentially licenses exactly one marker, so that the observed marker carries genuine information about the speaker's latent state.

The thresholds are ordered by marker strength:

$$\theta_{\text{sofern ich weiß}} < \theta_{\text{wie wir wissen}} < \theta_{ja} < \theta_{bekanntlich},$$

so that stronger markers require higher utility to be felicitous.

Upon observing u_i, the listener inverts this mapping:

$$P(pc_{\text{prop}}, pc_{\text{prag}}, g \mid u_i) \propto P(u_i \mid PC(pc_{\text{prop}}, pc_{\text{prag}}), g)\, P(pc_{\text{prop}})\, P(pc_{\text{prag}})\, P(g). \tag{4}$$

Finally, action adoption is modeled as a downstream decision based on the posterior:

$$P(\text{Adopt}(q) \mid u) = \iiint \sigma(\eta_0 + \eta_g g - \eta_{pc}\, PC(pc_{\text{prop}}, pc_{\text{prag}}))\, P(pc_{\text{prop}}, pc_{\text{prag}}, g \mid u)\, dpc_{\text{prop}}\, dpc_{\text{prag}}\, dg, \tag{5}$$

where σ is a logistic link function and $\eta_g, \eta_{pc} > 0$.

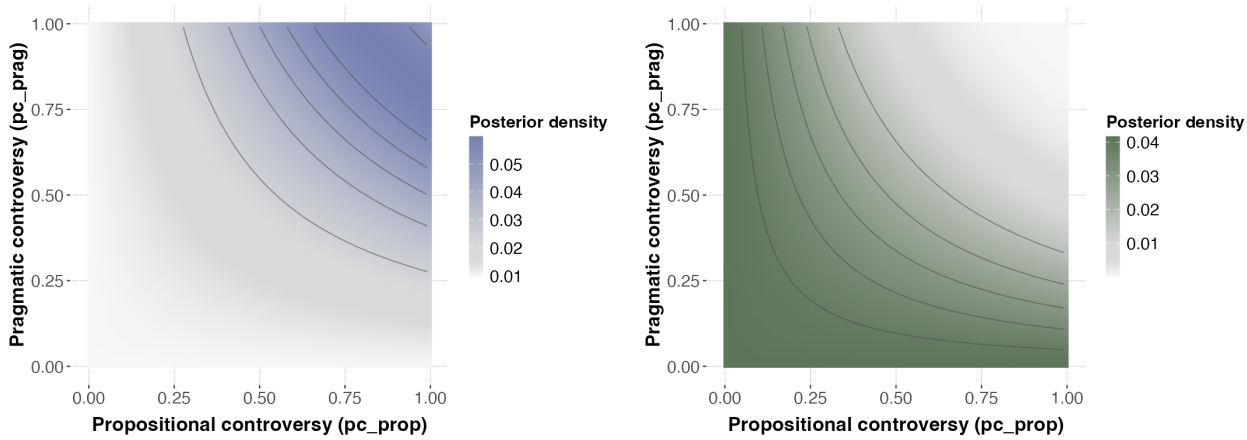
Model predictions (qualitative). The comprehension model predicts systematic inferences about perceived controversy and speaker goal strength from the observed marker u, and downstream effects on action adoption.

(1) Marker strength predicts lower overall perceived controversy. If markers lexicalize increasing thresholds

$$\theta_{\text{sofern ich weiß}} < \theta_{\text{wie wir wissen}} < \theta_{ja} < \theta_{bekanntlich},$$

then stronger markers license higher utility intervals and therefore support lower inferred overall controversy:

$$\mathbb{E}[pc \mid u = \textit{bekanntlich}] < \mathbb{E}[pc \mid u = \textit{ja}] < \mathbb{E}[pc \mid u = \textit{wie wir wissen}] < \mathbb{E}[pc \mid u = \textit{sofern ich weiß}].$$



(a) TBD

(b) TBD

Figure 2: TBD

In particular, stronger markers bias the listener towards interpretations in which p is sufficiently settled in the relevant community.

(2) Marker strength predicts higher inferred persuasive goal strength. Because stronger markers are licensed only when utility is high, and utility increases with g, observing a stronger marker shifts posterior mass toward higher g:

$$\mathbb{E}[g \mid u = \textit{bekanntlich}] > \mathbb{E}[g \mid u = \textit{ja}] > \mathbb{E}[g \mid u = \textit{wie wir wissen}] > \mathbb{E}[g \mid u = \textit{sofern ich weiß}].$$

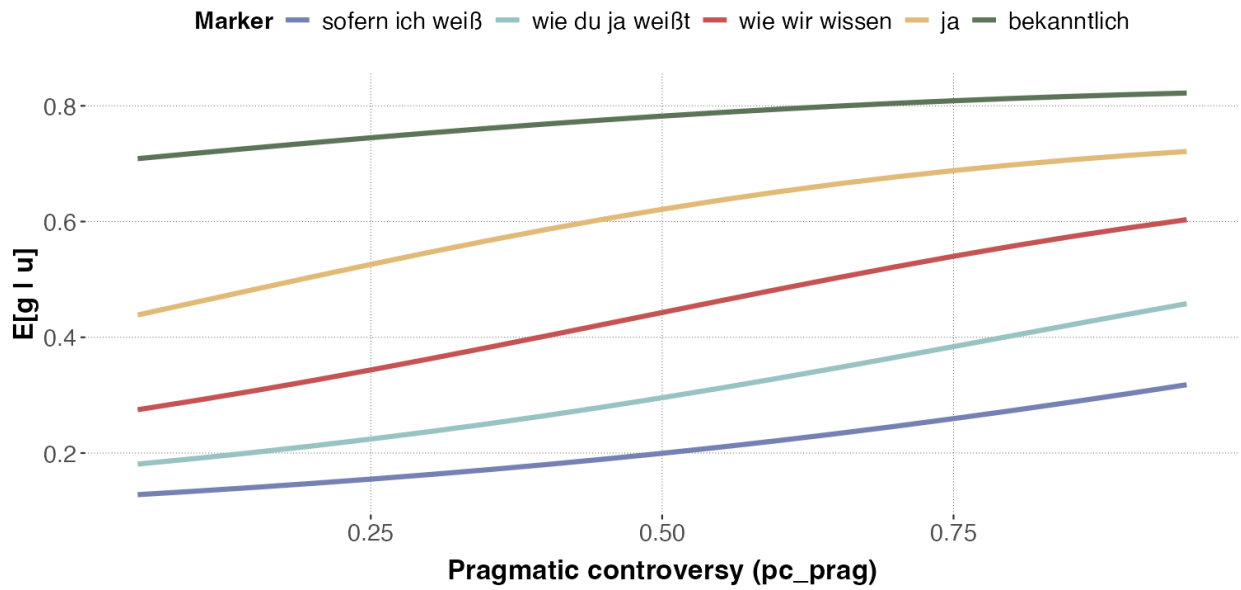
Thus, consensus markers trigger theory-of-mind inferences about the speaker's intention to push the addressee toward adopting q.

(3) Decomposition: propositional vs. pragmatic controversy. Since overall controversy is computed by

$$pc = PC(pc_{\text{prop}}, pc_{\text{prag}}) = pc_{\text{prop}} \cdot pc_{\text{prag}},$$

the model predicts a characteristic compensation pattern: high propositional controversy can be "rescued" by low pragmatic controversy (high community alignment), yielding low overall pc. Concretely, holding pc_{prop} fixed, stronger markers push inferences toward lower pc_{prag} (greater presumed alignment), whereas holding pc_{prag} fixed, stronger markers push inferences toward lower pc_{prop} (greater plausibility). This yields the prediction that the same marker can be licensed either by factual plausibility or by strong community alignment, and that listeners will trade off these sources of support in interpreting u.

(4) Interaction between marker strength and controversy in inferred goal strength. Because the utility includes an interaction term $-(pc \cdot g)$, the effect of controversy on inferred goal strength is super-additive: under higher inferred pc, observing a strong marker forces a stronger increase in inferred g in order to place the utterance within the marker's licensed utility interval. Thus, the



(a) TBD

Figure 3: TBD

model predicts the largest shifts in $\mathbb{E}[g \mid u]$ for strong markers in contexts where p is (or would ordinarily be) controversial.

(5) Downstream predictions for action adoption. Since adoption increases with g and decreases with pc,

$$P(\text{Adopt}(q) \mid u)$$

is predicted to be highest for strong markers when overall controversy is low, and reduced when overall controversy is high. Moreover, because marker strength jointly affects inferred pc and g, the model predicts (i) higher adoption for stronger markers in low-controversy contexts, and (ii) a characteristic dissociation under high controversy: strong markers may still increase perceived persuasive pressure (higher g) even when willingness to act remains limited by high pc.

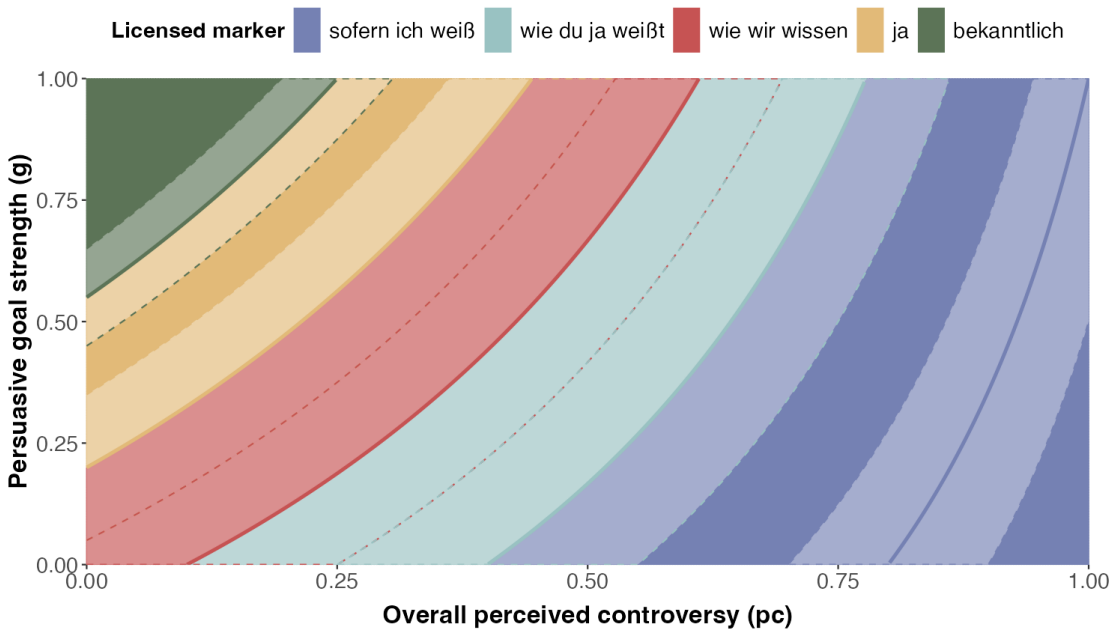
Experiments

We test the comprehension–production connection implied by the model: (i) listeners infer perceived controversy and speaker goal strength from a consensus marker and use these in downstream action decisions; (ii) speakers choose markers as a function of inferred controversy and persuasive goal strength.

0.1 General materials and factors

Markers. The utterance u contains one of five consensus markers (German):

$$u \in \{\textit{sofern ich weiß},\ \textit{wie du ja weißt},\ \textit{wie wir wissen},\ \textit{ja},\ \textit{bekanntlich}\},$$



(a) TBD

Figure 4: TBD

assumed to lexicalize increasing thresholds θ_u.

Proposition and action. Each item consists of a proposition p (an informational claim) and a proposed action q (a concrete recommendation). The target sentence frame is:

$$\text{MARKER}(p) \quad \Rightarrow \quad \text{therefore adopt } q.$$

Example (schematic):

> *Bekanntlich* p ("the new exam format yields more failing grades"). Therefore, q ("we should postpone the exam reform").

Manipulating controversy. We orthogonally manipulate two sources of perceived controversy:

- **Propositional controversy** (pc_{prop}): plausibility/compatibility with world knowledge and evidence strength (e.g. strong vs. weak evidence; plausible vs. surprising claim).
- **Pragmatic controversy** (pc_{prag}): expected disagreement / common-ground alignment in a specified community (e.g. shared expert consensus vs. polarized community).

Each item is paired with short contextual information implementing a 2×2 manipulation:

$$pc_{\text{prop}} \in \{\text{low}, \text{high}\} \quad \times \quad pc_{\text{prag}} \in \{\text{low}, \text{high}\}.$$

This directly targets the model decomposition $pc = PC(pc_{\text{prop}}, pc_{\text{prag}}) = pc_{\text{prop}} \cdot pc_{\text{prag}}$.

Pretests (recommended). We pretest contexts to validate that the manipulations selectively affect (i) plausibility/evidence (pc_{prop}) and (ii) expected disagreement/community alignment (pc_{prag}), using separate rating questions (see below).

0.2 Experiment 1: Comprehension (inference + adoption)

Goal. Test the listener-side predictions: (i) marker strength drives inferences about pc and g; (ii) decomposition trade-off between pc_{prop} and pc_{prag}; (iii) downstream effects on $\Pr(\text{Adopt}(q) \mid u)$.

Design. A mixed factorial design:

- **Within-item:** marker u (4 levels).
- **Within-item:** contextual manipulations pc_{prop} (2) and pc_{prag} (2).
- **Random factors:** participant and item (crossed).

We use a Latin-square scheme so each participant sees each *item* in exactly one marker condition (to avoid direct within-item comparisons of markers), while still seeing all factor levels across items.

Materials. Each trial shows:

1. a short vignette specifying an epistemic community C (e.g. "students in your department", "medical experts", "local residents"),
2. two context sentences implementing pc_{prop} and pc_{prag} (evidence strength vs. community alignment),
3. the target message: marker-applied claim p followed by recommended action q.

Procedure. For each trial, participants read the vignette and target message, then answer:

1. **Adoption decision (primary DV).**

$$\text{Would you adopt } q\text{? (yes/no)} \quad \text{or} \quad \text{slider } [0, 100].$$

2. **Posterior proxies (inference DVs).**
   - perceived overall controversy:

$$\widehat{pc} \text{ ("How controversial is } p \text{ in community } C\text{?")} \in [0, 100]$$

   - perceived speaker goal strength:

$$\widehat{g} \text{ ("How strongly is the speaker trying to persuade you to do } q\text{?")} \in [0, 100]$$

- decomposition checks (optional but strongly recommended):

$$\widehat{pc}_{\text{prop}}: \text{ “How plausible is } p \text{ given the evidence?”}$$

$$\widehat{pc}_{\text{prag}}: \text{ “How much disagreement would there be about } p \text{ in } C\text{?”}$$

3. **Attention/understanding checks** (item-specific multiple-choice).

Qualitative predictions. The model predicts:

1. **Marker strength → lower inferred controversy.** Holding context fixed,

   $\mathbb{E}[\widehat{pc} \mid u = \textit{bekanntlich}] < \mathbb{E}[\widehat{pc} \mid u = \textit{ja}] < \mathbb{E}[\widehat{pc} \mid u = \textit{wie wir wissen}] < \mathbb{E}[\widehat{pc} \mid u = \textit{sofern ich weiß}]$.

2. **Marker strength → higher inferred persuasive goal strength.**

   $\mathbb{E}[\widehat{g} \mid u = \textit{bekanntlich}] > \mathbb{E}[\widehat{g} \mid u = \textit{ja}] > \mathbb{E}[\widehat{g} \mid u = \textit{wie wir wissen}] > \mathbb{E}[\widehat{g} \mid u = \textit{sofern ich weiß}]$.

3. **Compensation pattern (decomposition).** Because $pc = pc_{\text{prop}} \cdot pc_{\text{prag}}$, a strong marker can be licensed either by low $\widehat{pc}_{\text{prop}}$ (high plausibility) or low $\widehat{pc}_{\text{prag}}$ (high alignment). Thus, effects of u should partially manifest as shifts in both components.

4. **Downstream adoption.** Adoption increases with $\widehat{g}$ and decreases with $\widehat{pc}$:

   $$\Pr(\text{Adopt}(q) \mid u) \text{ is higher for stronger markers in low-controversy contexts,}$$

   with a dissociation under high controversy: strong markers can increase $\widehat{g}$ while adoption remains suppressed by high $\widehat{pc}$.

Planned analysis (confirmatory). We estimate hierarchical regression models with crossed random effects (participant, item).

- **Inference models:**

$$\widehat{pc} \sim u * pc_{\text{prop}} * pc_{\text{prag}} + (1 + u \mid \text{participant}) + (1 + u \mid \text{item})$$

$$\widehat{g} \sim u * pc_{\text{prop}} * pc_{\text{prag}} + (1 + u \mid \text{participant}) + (1 + u \mid \text{item})$$

- **Adoption model:**

$$\Pr(\text{Adopt}(q)) \sim u * pc_{\text{prop}} * pc_{\text{prag}} + (1 + u \mid \text{participant}) + (1 + u \mid \text{item})$$

and (mediation-style) checks that $\widehat{pc}$ and $\widehat{g}$ statistically account for the marker effect:

$$\Pr(\text{Adopt}(q)) \sim u + \widehat{pc} + \widehat{g} + (1 \mid \text{participant}) + (1 \mid \text{item}).$$

0.3 Experiment 2: Production (marker choice under goals and controversy)

Goal. Test the speaker-side implications of the utility-threshold semantics: speakers select markers as a function of (i) persuasive goal strength g and (ii) perceived controversy pc, with particular sensitivity to the interaction $pc \cdot g$.

Design. A production-choice task with manipulated communicative goals and context:

- **Goal strength** g (2–3 levels; manipulated via incentives/instructions):

$$g \in \{\text{low (inform)},\ \text{medium (recommend)},\ \text{high (persuade/pressure)}\}.$$

- **Contextual controversy:** the same 2×2 manipulation of pc_{prop} and pc_{prag} as in Experiment 1.
- **Outcome:** marker choice u (4-way categorical), plus optional free-text realization.

Materials. Each trial provides:

1. the vignette (community C) and evidence/alignment information (implementing pc_{prop}, pc_{prag}),
2. the content to communicate: the proposition p and the intended action recommendation q,
3. a role/goal instruction manipulating g.

Goal manipulation (examples).

- **Low** g **(informative):** "Your task is to neutrally inform the addressee about p. Do not try to influence their action."
- **High** g **(persuasive):** "Your task is to get the addressee to adopt q. You have limited time and need to be effective."

Optionally, add an incentive-compatible bonus tied to the addressee's (simulated or real) adoption to operationalize g.

Procedure. On each trial, participants construct the message by selecting a marker from a drop-down menu:

$$[\text{marker}] + p \quad \text{then} \quad \text{recommend } q.$$

Optionally, participants can also edit the full sentence to allow naturalistic production, while the marker choice remains the main DV.

After producing the message, participants provide manipulation checks:

$$\widehat{pc}_{\text{prop}},\ \widehat{pc}_{\text{prag}},\ \widehat{g}$$

(brief slider questions to verify that the intended goal and context were perceived as such).

Predictions. Let marker strength correspond monotonically to θ_u.

1. **Goal strength increases marker strength.** Higher instructed g yields a higher probability of selecting strong markers:

$$\Pr(u = \textit{bekanntlich} \text{ or } \textit{ja} \mid g = \text{high}) > \Pr(u = \textit{bekanntlich} \text{ or } \textit{ja} \mid g = \text{low}).$$

2. **Controversy decreases marker strength.** Higher pc reduces the use of strong markers, due to the cost term in $U(p; pc, g)$.

3. **Interaction: controversy × goal.** Because U contains $-(pc \cdot g)$, the effect of g on marker choice is attenuated when pc is high: strong markers are most favored when g is high *and* pc is low; conversely, when pc is high, even high g yields more conservative markers.

4. **Decomposition trade-off.** High pc_{prop} can be compensated by low pc_{prag} (and vice versa). Thus, strong markers should be more likely in mixed cases (high/low) than in high/high.

Planned analysis. We model marker choice with an ordinal or multinomial mixed-effects model.

- **Ordinal approach:** map markers to an ordered scale by assumed θ_u and fit:

$$u_{\text{strength}} \sim g * pc_{\text{prop}} * pc_{\text{prag}} + (1 \mid \text{participant}) + (1 \mid \text{item}).$$

- **Multinomial approach:** directly predict categorical u:

$$\Pr(u = k) = \text{softmax}_k(\beta_{0k} + \beta_{gk} g + \beta_{pk} pc_{\text{prop}} + \beta_{rk} pc_{\text{prag}} + \cdots),$$

  with participant/item random intercepts (and slopes where identifiable).

Linking production and comprehension. A key cross-experiment test is whether the production pattern aligns with the comprehension-inferred thresholds: the same ordering of marker strength that drives lower inferred $\widehat{pc}$ and higher inferred $\widehat{g}$ in Experiment 1 should also be selected more frequently when speakers face low controversy and higher persuasive goals in Experiment 2.

Optional extension: closing the loop. To directly validate the downstream decision component, we can run a *yoked design*: messages produced in Experiment 2 are shown to new participants in a comprehension/adoption task (Experiment 1), testing whether naturally produced marker choices causally shift $\widehat{pc}$, $\widehat{g}$, and $\Pr(\text{Adopt}(q))$ as predicted.

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